



← Go Back

De

Language Reference

Funktionen >

Variablen >

Konstanten

Floating Point Constants

HIGH | LOW

INPUT | INPUT_PULLUP | OUTPUT

Integer Constants

LED_BUILTIN

true | false

Datentypen

array

bool

boolean

byte

char

double

float

int

long

short

size_t

string

Home / Programming / Language
Reference / Variablen / **INPUT |
INPUT_PULLUP | OUTPUT**

ON THIS PAGE

Defining Di

INPUT

INPUT_PUL

OUTPUT

See also

INPUT | INPUT_PULLUP | OUTPUT

Last revision ♦ 20.05.2024

Note: This article does not
exist in this language. If
you want to help us
translate it, you can find
the source code [here](#).

Defining Digital Pins modes

Digital pins can be used as
`INPUT`, `INPUT_PULLUP`, or
`OUTPUT`.

Changing a pin with `pinMode()`
changes the electrical behavior
of the pin.

INPUT

Arduino (ATmega) pins
configured as `INPUT` with

Help

`pinMode()` are said to be in a *high-impedance* state. Pins

configured as `INPUT` make extremely small demands on the circuit that they are sampling, equivalent to a series resistor of 100 Megohms in front of the pin. This makes them useful for reading a sensor.

If you have your pin configured as an `INPUT`, and are reading a switch, when the switch is in the open state the input pin will be "floating", resulting in unpredictable results. In order to assure a proper reading when the switch is open, a pull-up or pull-down resistor must be used. The purpose of this resistor is to pull the pin to a known state when the switch is open. A 10 K ohm resistor is usually chosen, as it is a low enough value to reliably prevent a floating input, and at the same time a high enough value to not draw too much current when the switch is closed. See the [Digital Read Serial](#) tutorial for more information.

If a pull-down resistor is used, the input pin will be `LOW` when the switch is open and `HIGH` when the switch is closed.

If a pull-up resistor is used, the input pin will be `HIGH` when the

INPUT_PULLUP

The ATmega microcontroller on the Arduino has internal pull-up resistors (resistors that connect to power internally) that you can access. If you prefer to use these instead of external pull-up resistors, you can use the `INPUT_PULLUP` argument in `pinMode()`.

See the [Input Pullup Serial](#) tutorial for an example of this in use.

Pins configured as inputs with either `INPUT` or `INPUT_PULLUP` can be damaged or destroyed if they are connected to voltages below ground (negative voltages) or above the positive power rail (5V or 3V).

OUTPUT

Pins configured as `OUTPUT` with `pinMode()` are said to be in a *low-impedance* state. This means that they can provide a substantial amount of current to other circuits. ATmega pins can source (provide current) or sink (absorb current) up to 40 mA (milliamps) of current to other

devices/circuits. This makes them useful for powering LEDs

because LEDs typically use less than 40 mA. Loads greater than 40 mA (e.g. motors) will require a transistor or other interface circuitry.

Pins configured as outputs can be damaged or destroyed if they are connected to either the ground or positive power rails.

See also

[Floating Point Constants](#)

[HIGH | LOW](#)

[Integer Constants](#)

[LED_BUILTIN](#)

[true | false](#)

Suggest changes	Need support?	License & Trademar
The content on docs.arduino.cc is facilitated through a public GitHub repository. If you see anything wrong, you can edit this page here.	Help Center Ask the Arduino Forum Discover Arduino Discord	The Arduino documentatic is licensed under the Creative Commons Attribution-Share Alike 4.0 license. Arduino and the Arduino logo are trademarks or registered trademarks of Arduino S.r.l.

Was this article helpful?

